Farhan Kumar

FPGA Design Engineer

RESUME_1722

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Total Experience: 4 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 4 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Hardware Simulation, Tcl, VHDL, UART, Intel Quartus, AXI4, Xilinx Vivado, Verilog, SPI, ILA, FPGA, DSP

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

FPGA Design Engineer | SparrowCloud Labs | Jan 2024 - Present

- Improved maximum operating frequency by 17% after timing-path optimization
- Reduced FPGA resource utilization by 38% through arithmetic sharing
- Lowered end-to-end data-path latency by 36% using pipelined logic
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate FPGA Design Engineer | Asterline Engineering | Jul 2022 - Dec 2023

- Lowered end-to-end data-path latency by 33% using pipelined logic
- Verified 604 protocol corner cases with repeatable simulation suites

PROJECTS

High-Speed Data Logger | FPGA, Xilinx Vivado, Verilog, ILA

- Created DMA and buffering logic with ILA debugging hooks.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

Protocol Adapter Core | Hardware Simulation, Verilog, ILA, FPGA

- Developed reusable SPI and UART modules with loopback tests.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Aravali Technical Institute, 2022
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

FPGA Architecture Workshop (2022)